

# The Drainage System — Stacks, Slope and Cleanouts

Waste moves by gravity alone. Every rule in this lesson is about keeping it moving at the right speed.

|                            |                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Soil or waste stack</b> | The vertical pipes into which waste flows from the drainage pipes.                                                                        |
| <b>Building drain</b>      | The horizontal run inside the building that collects from the stacks.                                                                     |
| <b>Building sewer</b>      | The run from the building out to the main or the septic system.                                                                           |
| <b>Cleanout</b>            | A drain coupling fitted with a removable plug so obstructions can be cleared. It provides access for service — that is its whole purpose. |
| <b>Backwater valve</b>     | A check valve that prevents public sewage flowing back into the house.                                                                    |
| <b>Cesspool</b>            | A pit in which raw sewage collects.                                                                                                       |

## The numbers here

|                                                                                                     |                               |
|-----------------------------------------------------------------------------------------------------|-------------------------------|
| Preferred slope on a waste pipe                                                                     | <b>1/4 to 1/2 in per foot</b> |
| Building sewer cleanout — within this distance of the junction of building drain and building sewer | <b>10 ft</b>                  |

The slope range matters in both directions. Too flat and the solids do not carry; too steep and the water runs away from the solids and leaves them behind. A quarter to a half inch per foot keeps liquid and solid traveling together.

## YOU HAVE MISSED THIS

**The waste line clean-out provides:**  
**Access for service.**

## YOU HAVE MISSED THIS

**What is the reason for the square shape of a plumbing drain cleanout?**  
**To avoid the possibility of stripping the plug during removal.**

A square boss gives a wrench four flat faces to grip. A round or shallow plug rounds off under force, and a cleanout is by definition something you are opening because the drain is already blocked.

## Drainage has no minimum pressure

Asked for the minimum allowable pressure in the plumbing drainage system, the answer is none of them. Drainage is a gravity system — pressure belongs to the supply side. If a question about drainage offers you a pressure, the offer itself is the trap.

## Say it back

Cover the page. Answer each one out loud.

1. What is a soil or waste stack?
2. Give the waste pipe slope range, and say what goes wrong at each extreme.
3. What does a cleanout provide, and why is its plug square?
4. How far from the drain-to-sewer junction must the building sewer cleanout sit?
5. What does a backwater valve stop, and what is a cesspool?
6. What is the minimum pressure for the drainage system?